

HRMS

Human Resource Management System

Solution Design Document

Custom ERP Application — Built from Scratch

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1. Executive Summary

This document describes the complete solution design for a custom-built Human Resource Management System (HRMS). The system is being developed as a standalone custom ERP application, built from scratch, to meet organizational requirements that go beyond the capabilities of standard off-the-shelf HRMS platforms.

The core philosophy of this HRMS is to move from a time-based evaluation model to a value-based model — measuring employees by their professional capability, work quality, efficiency, asset ownership, and long-term contribution to the organization.

This document covers the project scope, functional module design, business logic specifications, data ownership framework, reward model, and high-level non-functional requirements. It is intended for use by stakeholders, project managers, architects, developers, and QA teams.

2. Project Scope

2.1 In Scope

The following capabilities are within the scope of this project:

- Base Professional Value (BPV) Engine — a configurable scoring engine to calculate employee professional worth
- Outcome-Based Time Reporting — replacing traditional 8-hour attendance with delivery-based reporting
- Dual-Component Salary Structure — BPV-driven base pay + small time-reporting variable component
- Efficiency Evaluation and Scoring — rewarding fast, high-quality task completion
- Employee Data Ownership Registry — tracking every digital asset with full ownership metadata
- Documentation Compliance Management — enforcing and tracking documentation requirements on all assets
- Long-Term Reward and Royalty Management — configurable incentive engine tied to asset value creation
- Beneficiary Management — employee-designated beneficiaries for long-term reward distribution
- HR and Finance Administration Dashboard — configuration, reporting, and oversight
- Reports and Analytics — productivity, efficiency, contribution, and reward analytics

2.2 Out of Scope

The following are explicitly out of scope for the initial release:

- Payroll disbursement and banking integrations
- Recruitment and applicant tracking
- Leave management and travel expense management
- Performance appraisal ratings (handled by Efficiency Score, not separate appraisal cycles)
- Third-party HRMS integrations (Salesforce, SAP, Workday, etc.)
- Mobile application (web-based responsive UI is in scope; native mobile app is a future phase)

2.3 Assumptions

- The application will be hosted on a cloud platform (provider to be confirmed).
- Authentication will be handled via role-based access control (RBAC) with SSO support.
- All monetary calculations are configurable and do not hardcode any currency or rate.
- Legal and compliance requirements for beneficiary payouts will be handled by HR/Finance configuration, not system enforcement.

3. System Overview

The HRMS is a web-based custom ERP application with a modular architecture. Each module is independently configurable by HR and Finance administrators. The system is designed to scale with organizational growth and adapt to changing policies without code changes.

3.1 High-Level Module Map

Module	Purpose
Employee Master	Central employee profile and professional data management
BPV Engine	Calculate and maintain each employee's Base Professional Value
Qualification & Experience Scoring	Input scoring for education, experience, and org profile
Technical Skill & Certification Management	Manage skills, certifications, and proficiency ratings
Outcome-Based Task Reporting	Record assigned tasks, delivered work, and quality assessments
Time Reporting	Minimal time logging (1/2/4/6 hrs) linked to task completion
Dual-Component Salary Management	Manage BPV salary component and time-reporting variable component
Productivity & Efficiency Analytics	Score and trend employee efficiency over time
Data Ownership Registry	Register, track, and govern all employee-created digital assets
Asset & Documentation Management	Enforce documentation compliance for all owned assets
Contribution Tracking	Link employee contributions to business outcomes
Reward & Incentive Management	Configure and calculate long-term rewards and royalties
Beneficiary Management	Employee beneficiary nomination and management
HR & Finance Admin Dashboard	System-wide configuration, oversight, and approvals
Reports & Analytics	Cross-module reporting and business intelligence

4. Functional Module Specifications

4.1 Employee Master

The Employee Master is the central profile for each employee. All other modules reference this record.

Key Data Fields

- Personal information: name, date of birth, contact details, emergency contact
- Employment details: employee ID, designation, department, reporting manager, date of joining
- Professional profile: qualifications, certifications, experience history, organization history
- Skill profile: technical skills, proficiency levels, tools, platforms, frameworks
- BPV Score: computed and maintained by the BPV Engine
- Salary components: BPV Salary (Component 1) and Time Reporting Value (Component 2)
- Status: active, on notice, exited (ownership records are retained permanently)

4.2 Base Professional Value (BPV) Engine

The BPV Engine is the core differentiator of this HRMS. It calculates a numeric score representing the professional worth of each employee, which directly drives the majority of their compensation.

4.2.1 BPV Formula

BPV Score = Education Score + Experience Score + Organization Score + Technical Skill Score + Certification Score

Each factor is independently weighted. Weights are configurable by HR administrators through the Admin Dashboard. The total weight must always sum to 100%.

4.2.2 Education Scoring

Qualification	Default Weight Contribution
12th Grade	Base level
Diploma	Incremental above 12th
Undergraduate Degree (3-Year)	Standard undergraduate level
Undergraduate Degree (4-Year)	Engineering / professional level
Master's Degree	Postgraduate level
Double Master's Degree	Dual specialization
MBA	Management specialization
M.Tech / MS	Technical postgraduate
PhD	Doctoral research level
Post-Doctoral Fellowship	Advanced research recognition

4.2.3 Experience Scoring

- Total years of professional experience

- Domain-specific experience (years in the relevant domain)
- Enterprise-level experience
- Global organization experience
- Leadership and management experience

4.2.4 Organization Profile Scoring

The type of organization an employee has worked at contributes to their BPV score:

- Startup
- Mid-size organization
- Enterprise organization
- Fortune 500 company
- Government organization
- Research organization

4.2.5 Technical Skill Scoring

- Programming languages (each with a proficiency rating: beginner / intermediate / expert)
- Platforms and cloud environments
- Frameworks and libraries
- Tools and methodologies (Agile, DevOps, etc.)
- Industry certifications (AWS, Azure, PMP, CPA, etc.)

4.2.6 BPV Recalculation Rules

- BPV is recalculated automatically when an employee's profile is updated.
- Manual override is permitted by HR Admin with a mandatory reason log.
- BPV history is retained — showing score at each point in time.
- Salary Component 1 is updated based on the new BPV score at each review cycle.

4.3 Outcome-Based Task and Time Reporting

This module replaces the traditional attendance-and-timesheet model. The focus is on what was delivered, not how many hours were occupied.

4.3.1 Task Reporting

- Managers assign tasks to employees with defined deliverables, quality criteria, and expected completion dates.
- Employees report task completion with: completion status, quality assessment (self), any blockers or dependencies, and time consumed.
- Managers or reviewers confirm delivery quality and mark tasks as accepted or returned for revision.

4.3.2 Time Reporting Options

Employees select the time block that reflects actual time spent completing the assigned work:

Time Block	Usage Condition
1 hour	Very short, well-scoped tasks completed in full
2 hours	Short tasks completed to required quality
4 hours	Half-day equivalent work completed
6 hours	Near-full day of work completed

Employees are not required to report 8 hours if the assigned work has been completed. Artificial inflation of time logs is discouraged and tracked.

4.3.3 Quality Assessment

- Quality is rated by the reviewing manager on a defined scale.
- Quality scores feed directly into the employee's Efficiency Score.
- Low quality scores trigger a review flag — visible to HR.

4.4 Dual-Component Salary Structure

The compensation model for each employee consists of two distinct components. This separation ensures that professional expertise is recognized independent of daily time reporting.

Component 1 — Base Professional Value Salary (Majority)

- Constitutes the majority of total compensation (percentage configurable by HR/Finance).
- Calculated directly from the employee's BPV Score.
- Reviewed and updated at each performance review cycle.
- Independent of daily time reporting.
- Reflects the employee's academic qualifications, experience, skills, and market value.

Component 2 — Time Reporting Value (Minority)

- A small variable component — compensates for operational work tracking.
- Computed based on time entries submitted and accepted for the reporting period.
- Does not dominate overall compensation.
- Designed to cover administrative overhead, not to measure performance.

Configuration

- HR and Finance administrators configure the Component 1 / Component 2 split ratio.
- The split ratio can differ by employee grade, department, or designation.
- All configurations are version-controlled with effective dates.

4.5 Efficiency Evaluation

The Efficiency Score tracks how effectively an employee delivers work relative to time consumed and quality achieved. This score is a key driver of annual performance incentives.

4.5.1 Efficiency Score Calculation

Efficiency Score = f(Task Completion Rate, Quality Score, Time Consumed vs. Benchmark)

- Task Completion Rate: percentage of assigned tasks delivered on time.
- Quality Score: average quality rating received from reviewers.
- Time Efficiency: comparison of time reported vs. benchmark time for similar tasks.

4.5.2 Example Scenarios

Scenario	Outcome
Employee completes task in 2 hrs, high quality	High efficiency score → higher annual incentive
Employee completes same task in 6 hrs, same quality	Lower efficiency score → lower incentive
Employee completes task in 2 hrs, poor quality, revision needed	Quality penalty applied → neutral or low score
Employee consistently delivers under benchmark with high quality	Top efficiency tier → maximum incentive eligibility

The system does not penalize an employee simply for reporting fewer hours. It rewards quality delivery at reduced time. The combination of low time + high quality = best outcome.

4.6 Employee Data Ownership Registry

Every digital asset created or contributed to by an employee must be registered in the Data Ownership Registry with complete ownership metadata. This creates a permanent, traceable record of all organizational knowledge and intellectual property.

4.6.1 Asset Types Covered

- Source code and code modules
- Technical documents and specifications
- Design files and wireframes
- Reports and dashboards
- Data models and database schemas
- Automation scripts and pipelines
- Knowledge articles and wikis
- Architecture documents and ADRs
- AI prompts and prompt libraries
- Configuration files and environment definitions

4.6.2 Ownership Metadata per Asset

Field	Description
Asset ID	Unique system-generated identifier
Asset Name	Human-readable name of the asset
Asset Type	Category from the defined asset type list
Creator	Employee who originally created the asset
Contributors	All employees who contributed to the asset
Reviewer	Employee who reviewed the asset
Approver	Employee or manager who approved the asset
Current Owner	Active responsible owner
Creation Date	Date of first creation
Last Modified	Date of most recent change
Version	Current version number
Version History	Full changelog with date and contributor per version
Project / Department	Associated project or department
Status	Active / Archived / Deprecated

4.6.3 Post-Departure Ownership Rule

When an employee exits the organization:

- Operational ownership of all their assets transfers to the organization.
- Historical authorship records remain permanently attributed to the individual.
- The system flags unhandled ownership transfers for HR review upon exit.
- A handover checklist is generated to ensure all assets are documented and transferred.

4.7 Documentation Compliance Management

Every registered asset must have complete documentation before it is considered compliant. Incomplete documentation is flagged automatically.

4.7.1 Required Documentation per Asset

- Implementation guide — how the asset was built or configured
- User guide — how to use or operate the asset
- Maintenance instructions — how to maintain, update, or fix the asset
- Dependencies — list of tools, services, libraries the asset depends on
- Version history — a log of all changes with dates and authors

4.7.2 Compliance Enforcement

- The system calculates a Documentation Compliance Score (0–100%) per asset.
- Assets with compliance below a configured threshold are flagged as non-compliant.
- Non-compliant assets appear in the HR/Finance admin dashboard.
- Employee's contribution score may be affected by documentation compliance of their owned assets.

4.8 Long-Term Reward and Royalty Management

The Reward Engine enables the organization to provide ongoing incentives to employees whose created assets continue to generate measurable business value.

4.8.1 Reward Basis Metrics

Rewards are calculated based on the following value indicators (all configurable):

- Revenue generated by or attributable to the asset
- Operational cost savings produced by the asset
- Productivity improvements measured from asset adoption
- Customer adoption or usage metrics
- Intellectual property value assigned to the asset
- Strategic business impact as assessed by management

4.8.2 Reward Types

Reward Type	Description
Annual Bonus	One-time yearly payment linked to asset performance
Innovation Incentive	Award for assets with measurable innovation value
Profit Sharing	Percentage of profit attributable to the asset
Royalty-Based Reward	Ongoing periodic payment tied to continued asset use or revenue
Long-Term Recognition Award	Non-monetary or hybrid recognition for multi-year value creation

4.8.3 Reward Calculation

- The calculation method and formula are fully configurable by HR and Finance administrators.
- Multiple reward types can apply to a single asset simultaneously.
- All reward calculations are logged with version history and effective dates.
- Rewards are linked to both the Creator and Contributors proportionally (configurable split).

4.9 Beneficiary Management

Employees may nominate one or more beneficiaries to receive eligible long-term rewards in their place, subject to organizational policy and applicable legal requirements.

4.9.1 Eligible Beneficiary Types

- Spouse
- Children
- Other family members or legal heirs
- Charitable organizations
- Non-governmental organizations (NGOs)
- Educational institutions
- Orphanages
- Registered trusts

4.9.2 Beneficiary Record Fields

- Beneficiary name and relationship to employee
- Contact details
- Allocation percentage (if multiple beneficiaries)
- Supporting documentation (legal or identity verification as required)
- Designation date and effective date

4.9.3 Default Rule

If no beneficiary is designated at the time a reward becomes payable, the system routes the reward handling to HR for resolution per organizational policy and applicable legal requirements.

5. Non-Functional Requirements

5.1 Security

- Role-Based Access Control (RBAC) — distinct roles for Employee, Manager, HR Admin, Finance Admin, System Admin.
- All sensitive data (salary, BPV scores, asset ownership) is accessible only to authorized roles.
- Audit trail for all data changes — who changed what and when.
- SSO support for enterprise authentication.

5.2 Performance

- BPV score recalculation must complete within 5 seconds for a single employee profile update.
- Dashboard and analytics pages must load within 3 seconds under normal load.
- System must support concurrent use by up to 500 users without degradation.

5.3 Scalability

- Architecture must support horizontal scaling as the organization grows.
- Asset Ownership Registry must support up to 1 million asset records without query degradation.

5.4 Configurability

- All scoring weights, salary split ratios, reward formulas, and compliance thresholds must be admin-configurable without code deployment.
- Configuration changes must be versioned with effective dates and rollback capability.

5.5 Data Integrity

- All ownership records, BPV histories, and reward calculations must be immutable — append-only logs.
- No permanent deletion of ownership or reward records — only archival.

6. User Roles and Access

Role	Key Permissions
Employee	View own profile, submit task reports, log time, register assets, nominate beneficiaries
Manager	Assign tasks, review and accept deliveries, rate quality, view team efficiency scores
HR Admin	Configure BPV weights, manage salary structure, manage reward rules, view all employee profiles
Finance Admin	Configure salary components, review reward calculations, manage beneficiary payouts
System Admin	Full system configuration, user management, audit log access

7. Expected Business Benefits

- Rewards professional expertise and knowledge rather than time spent.
- Encourages faster, high-quality task completion through Efficiency Scoring.
- Aligns compensation with real professional value and market worth via BPV.
- Improves accountability for all organizational digital assets through the Ownership Registry.
- Ensures knowledge and authorship are never lost — even after an employee exits.
- Enforces documentation discipline, reducing knowledge debt and maintenance risk.
- Strengthens long-term employee engagement through configurable, asset-linked rewards.
- Aligns employee rewards directly with the business value they create.

8. Document Control

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